MATERIAL SAFETY DATA SHEET

VESUVIUS

APR 2 4 2007

Vesuvius Safety Data Sheet.

1. PRODUCT AND COMPANY INFORMATION

Product Name:	**TRIAD® 60**
UN Number	No UN Number Allocated
Product Type:	Monolithic Refractory (Castable)
TS Number:	LEC30277

Company Name And Address:	Vesuvius USA 495 Emma Street Bettsville, OH 44815
Technical Contact:	Phone #: 1-419-986-5126
Emergency Contact:	24hr. Emergency Assistance (CHEMTREC) 1-800-424-9300 Outside the Continental U.S. See Section 15 or Call Chemtrec Collect: 703-527-3887

2. COMPOSITION / INFORMATION ON INGREDIENTS

Substance	% Range	Identification No.	Exposure Limits U.S.A.
Aluminium Oxide	7-14	CAS 1344-28-1	**Nuisance Particulate** ACGIH TLV:TWA (resp.) 10mg/m^3 OSHA PEL:TWA (resp.) 5mg/m^3
Aluminum Silicate	75-99	CAS 1302-93-8	**Nuisance Particulate**
Crystalline Silica, Cristobalite	0.1-1	CAS 14464-46-1	ACGIH TLV:TWA (resp.) 0.025mg/m^3 OSHA PEL:TWA (resp.) 10mg/m^3÷ (%SiO_2+2)
Silica, Amorphous	10-20	CAS 60676-86-0	ACGIH TLV: **Withdrawn**
Magnesium Oxide	1-5	CAS 1309-48-4	**Nuisance Particulate**
Silica, Amorphous	5-10	CAS 69012-64-2	ACGIH TLV: **Withdrawn**

3. HAZARDS IDENTIFICATION

Emergency Overview:	Mixtures of brown-gray, and white coarse to fine particles. Product is not combustible. Not a spill hazard.
Precautions:	Pre-existing lung conditions such as, but not limited to bronchitis, emphysema and asthma.
Chronic Health Effects:	Prolonged inhalation of product may lead to the development of a disabling pulmonary fibrosis known as silicosis, which may lead to cancer.
Acute Health Effects:	Eyes: Corrosive and physical eye irritant. Skin: Slight skin irritation Inhalation: Irritation of upper respiratory system Ingestion: May cause gastrointestinal disturbances

4. FIRST AID MEASURES

Inhalation:	Remove victim to fresh air. If not breathing, give artificial respiration and seek medical attention.
Eye contact:	Flush eyes with large amounts of water. Seek medical attention if irritation persists.
Skin contact:	Wash affected area with mild soap and water.
Ingestion:	Seek medical attention if symptoms persist.

5. FIRE FIGHTING MEASURES

Extinguisher Type:	No special instructions or conditions.
Special Procedures:	No special instructions other than use of approved respirators. Product is not a combustible.

6. ACCIDENTAL RELEASE MEASURES

Spillage:	No special requirements. Use safety glasses, gloves, skin protection, and respiratory protection.

7. HANDLING AND STORAGE

Handling	Use safety glasses, gloves, skin protection, and respiratory protection.
Storage	No special requirements.

8. EXPOSURE CONTROLS AND PERSONAL PROTECTION

Ventilation:	Provide sufficient ventilation, in both volume and airflow patterns, to control any mist/particulate emissions below allowable limits. See Exposure Limits in Section 2.
Respiratory:	Provide workers with legally approved respirators for level of exposure incurred.
Eye:	The use of proper eye protection is recommended (ex. safety glasses).
Hand:	The use of proper hand protection is recommended (ex. barrier cream with anti-slip gloves).
Other:	Safety shoes and long sleeve shirts are recommended for foot and skin protection.

9. PHYSICAL AND CHEMICAL PROPERTIES

Appearance: A mixture of black-gray, and white coarse to fine particles; plastic fibers may be present; odorless.

Boiling Point:	Not Applicable
Melting Point:	>2900°F (1590°C)
Bulk Density:	See specific Product Data Sheet
% Volatile by Volume:	0
Evaporation Rate:	Not applicable
Water Solubility:	<1%
pH (in 10% slurry):	9-10
Specific Gravity (g/cc):	Mixture

10. STABILITY AND REACTIVITY

Chemical Stability	Stable
Incompatibility	None
Hazardous Decomposition	None

11. TOXICOLOGICAL INFORMATION

Substance & CAS Number	Carcinogenic Data	Summary of Hazards
Aluminium Oxide (CAS 1344-28-1)	No Data Found	Respiratory: Nuisance Particulate
Aluminum Silicate (CAS 1302-93-8)	No Data Found	Respiratory: Nuisance Particulate
Magnesium Oxide (CAS 1309-48-4)	No Data Found	Respiratory: Nuisance Particulate
Silica, Fused (CAS 60676-86-0)	IARC Group 3	Respiratory: Inhalation Hazard
Silica, Fume (CAS 69012-64-2)	IARC Group 3	Respiratory: Inhalation Hazard
Cristobalite (CAS 14464-46-1)	IARC Group 1	Respiratory: May Develop Silicosis

12. ECOLOGICAL INFORMATION

No Data Available

13. DISPOSAL INFORMATION

This product, as manufactured, does not exhibit any characteristics of a hazardous waste. It is suitable for landfill disposal. However, debris generated during installation, maintenance or tear-out procedures may be contaminated with other hazardous materials. Therefore, appropriate waste analysis may be necessary to determine proper disposal. A qualified environmental professional in accordance with applicable federal, state, and local regulations should determine waste characterisation and disposal/treatment methods.

14. TRANSPORT INFORMATION

This product is not classified as a hazardous material for transportation. No hazard class, no label or placard required, no UN or NA number assigned. Shipment outside the U.S. should be reviewed by an environmental professional for country specific regulatory requirements.

15. REGULATORY INFORMATION

Products or components of mixture regulated under the following
U.S.A. Regulations:
Sara Title III: (302/304) No; (311/312) Yes; (313) No
CERCLA (RQ): No
TSCA: Yes (All Substances listed in ingredients)
California Proposition 65: Yes (Silica)
HMIS Codes: Health: 2 Fire: 0 Reactivity: 0 Protection: E
Canadian Regulations:
Domestic Substance List (DSL): Yes (All Substances Listed)
WHMIS Class: D2A;E
Contact Phone # 905-732-4441

16. OTHER INFORMATION

Removal After Service/Tear-Out Precautions:
Because of the possible presence of crystalline silica in used refractory debris, particular care should be exercised during tear-out to minimize the generation of dust. Adherence to proper methods of dust suppression and control is imperative. The following precautions should be taken during tear-out.
1. Employees should be apprised of the hazards and proper conditions and precautions for safe use or exposure.
2. Approved respirators, in accordance with requirements of 29 CFR 1910.134, should be used for dust levels above 0.05mg/m^3 respirable crystalline silica.
3. Dust generation should be minimized by the use of dust control equipment or water spray.
4. Wear protective clothing and vacuum clean prior to removing clothing.
5. Where there is a possibility of exposure to dust containing respirable crystalline silica, the following warning should be posted.

FREE SILICA WORK AREA	AVOID BREATHING DUST
DUST MAY CAUSE DELAYED LUNG INJURY(SILICOSIS)	

ACRONYMS AND REFERENCES USED IN PREPARATION OF MSDS':

ACGIH: American Conference of Governmental Industrial Hygienists
CAS#: Stands for Chemical Abstracts Service
CERCLA: Comprehensive Environmental Response, Compensation & Liability Act
IARC: International Agency for Research on Cancer
Group 1: Carcinogenic to Humans. (IARC)
Group 2A: Probably Carcinogenic to Humans. (IARC)
Group 2B: Possibly Carcinogenic to Humans. (IARC)
Group 3: Unclassifiable as to Carcinogenicity in Humans. (IARC)
Group 4: Probably not Carcinogenic to Humans. (IARC)
HMIS: Hazardous Materials Identification (National Paint and Coatings Association)
mg/m^3: Milligrams per cubic meter
OSHA: Occupational Safety and Health Administration
PEL: Permissible Exposure Limit (OSHA)
SARA: Superfund Amendments and Reauthorization Act
TITLE III: Emergency Planning and Community Right To Know Act
Section 302: Extremely Hazardous Substances
Section 304: Emergency Release
Section 311: *Community Right-to-Know*, MSDSs or List of Chemicals
Section 312: *Community Right-to-Know*, Inventories & Locations, (Tier I/Tier II)
Section 313: Toxic Chemicals, Toxic Chemical Release Reporting, Form R
TLV: Threshold Limit Values (ACGIH)
TWA: Time Weighted Average

REFERENCES:

Sax, N. Irving: Dangerous Properties of Industrial Materials, Ninth Edition, Van Nostrand Reinhold Co., Inc., 1996.
Kirk, R. and Othmer, D., Encyclopedia of Chemical Technology, Third Edition, Wiley-Interscience, New York, NY 1982.
Clansky, K.B., Suspect Chemicals Sourcebook, 1992-2 Edition, Roytech Publications, Bethesda, Maryland.
Sax, N.Irving and Lewis,R.J. Hawley's Condensed Chemical Dictionary, Eleventh Ed., Van Nostrand Reinhold Co.,Inc., NY
Manufacturers/Suppliers, Material Safety Data Sheets on Raw Materials Used
American National Standard for Hazardous Industrial Chemicals - Material Safety datasheets - Preparation, American National Standards Institute, Inc.11 West 42nd St, New York, NY 10036.

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